Tasnimul Alam Taz

PhD Candidate
Wayne State University
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Research Interests

Bioinformatics, Machine Learning, Generative AI, Explainable AI, Graph Representation Learning, Statistical Genomics, Precision Oncology, Biomarker Identification, Next-generation sequencing data analysis, Multi-omics data integration.

Education

Wayne State University, Michigan, USA

Doctor of Philosophy
Department of Computer Science, May 2027 (Anticipated)
Current GPA: 3.82/4.0
Advisor: Dr. Suzan Arslanturk
Dissertation title: Genomic Biomarker Discovery for Cancer Prognosis and Precision Oncology.

Wayne State University, Michigan, USA

Masters of Science
Major in Computer Science, December 2023
Advisor: Dr. Suzan Arslanturk
GPA: 3.78/4.0

Daffodil International University, Dhaka, Bangladesh

Bachelor of Science
Software Engineering, June 2021
GPA: 3.80/4.0

Publications

1. **Tasnimul Alam Taz**, Melike Yildirim, and Suzan Arslanturk. "CLOVER: A Cross-Cancer Learning Model Using Somatic Variant Data for Biomarker Recognition." In *Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (ACM-BCB 2026)*, Rende, Italy, June 30–July 3, 2026.
2. **Tasnimul Alam Taz** and Suzan Arslanturk. "Unveiling Genetic Variant-Level Biomarkers for Aggressive Prostate Cancer." *Informatics in Medicine Unlocked*, vol. 43, Article 101410, 2023.
3. Seeya Awadhut Munj, **Tasnimul Alam Taz**, Suzan Arslanturk, and Elisabeth I. Heath. "Biomarker-Driven Drug Repurposing on Biologically Similar Cancers with DNA-Repair Deficiencies." *Frontiers in Genetics*, vol. 13, Article 1015531, 2022.

4. **Tasnimul Alam Taz**, Kawsar Ahmed, Bikash Kumar Paul, Fahad Ahmed Al-Zahrani, S. M. Hasan Mahmud, and Mohammad Ali Moni. "Identification of Biomarkers and Pathways for SARS-CoV-2 Infections That Cause Complications in Pulmonary Arterial Hypertension Patients." *Briefings in Bioinformatics*, vol. 22, no. 2, pp. 1451–1465, 2021.
5. **Tasnimul Alam Taz**, Kawsar Ahmed, Bikash Kumar Paul, Md Kawsar, Nargis Aktar, S. M. Hasan Mahmud, and Mohammad Ali Moni. "Network-Based Identification Genetic Effect of SARS-CoV-2 Infections to Idiopathic Pulmonary Fibrosis (IPF) Patients." *Briefings in Bioinformatics*, vol. 22, no. 2, pp. 1254–1266, 2021.
6. **Tasnimul Alam Taz**, Md Kawsar, Sinthia Siddique, Kawsar Ahmed, Mohammad Ali Moni, and Bikash Kumar Paul. "Drug Compound Prediction-Based Analysis of Cigarette Smoking in Pancreatic Cancer Patients: A Bioinformatics Study." In *Proceedings of the 2020 IEEE International Women in Engineering Conference on Electrical and Computer Engineering (WIECON-ECE)*, pp. 320–323, IEEE, 2020.
7. **Tasnimul Alam Taz**, Md Kawsar, Bikash Kumar Paul, Kawsar Ahmed, and Touhid Bhuyian. "Characterizing Topological Properties and Network Pathway Models Among Vector-Borne Diseases." *Informatics in Medicine Unlocked*, vol. 18, Article 100312, 2020.
8. **Tasnimul Alam Taz**, Md Kawsar, Bikash Kumar Paul, and Kawsar Ahmed. "Computational Analysis of Regulatory Gene Network Pathways Among Devastating Cancer Diseases." *Journal of Proteins and Proteomics*, vol. 11, no. 1, pp. 63–76, 2020.
9. Md Kawsar, **Tasnimul Alam Taz**, Bikash Kumar Paul, Kawsar Ahmed, Md Ahsan Habib, and Touhid Bhuyian. "Identification of Vital Regulatory Genes with Network Pathways Among Huntington's, Parkinson's, and Alzheimer's Diseases." *Network Modeling Analysis in Health Informatics and Bioinformatics*, vol. 9, no. 1, Article 50, 2020.
10. Md Kawsar, **Tasnimul Alam Taz**, Bikash Kumar Paul, Shahin Mahmud, Md Manowarul Islam, Touhid Bhuyian, and Kawsar Ahmed. "Analysis of a Gene Network Model of Thyroid Disorder and Associated Diseases: A Bioinformatics Approach." *Informatics in Medicine Unlocked*, vol. 20, Article 100381, 2020.

Academic Employment

Graduate Teaching Assistant, May 2023 - August 2026

Department of Computer Science

Wayne State University, Detroit, MI

Graduate Research Assistant, August 2021 - April 2023

Department of Computer Science

Wayne State University, Detroit, MI

Project: Worked on DoD grant #W81XWH-21-1-0570 and NSF grant #1948338 under the supervision of Dr. Suzan Arslanturk on developing statistical and AI-based methods for genomic biomarker discovery. The research focused on identifying biomarkers of aggressive cancers, integrating multi-cancer genomic data, and supporting precision oncology through explainable machine learning and drug repurposing.

Grant Writing Experience

DoD PCRP grant: Multi-task learning system for single-cell biomarker discovery for Prostate cancer. PI: Dr. Suzan Arslanturk, Graduate Researcher: **Tasnimul Alam Taz**, (Award amount: $1 million,

Under submission). As a graduate researcher, I contributed to defining the project's specific aims, identifying relevant data sources, conducting preliminary experiments, and assisting with the preparation and writing of the grant proposal.

Teaching Experience

Course Instructor (Wayne State University, MI, USA)

- CSC 2110: Computer Science I (Object-oriented programming): Spring/Summer 2026
- CSC 2110: Computer Science I (Object-oriented programming): Spring/Summer 2024
- CSC 2110: Computer Science I (Object-oriented programming): Spring/Summer 2023

Teaching Assistant (Wayne State University, MI, USA)

- CSC 4996: Senior Capstone Project: Winter 2026
- CSC 5800: Intelligent Systems: Algorithms and Tools: Fall 2025
- CSC 4996: Senior Capstone Project: Spring/Summer 2025
- CSC 4996: Senior Capstone Project: Winter 2025
- CSC 4996: Senior Capstone Project: Fall 2024
- CSC 4996: Senior Capstone Project: Winter 2024

Undergraduate Research Experience

Research Assistant

Department of Software Engineering

Daffodil International University, Dhaka, Bangladesh

January 2019 – June 2021

Conducted bioinformatics research on cancer and infectious diseases using gene-expression analysis, protein–protein interaction networks, pathway analysis, and computational drug prediction under the supervision of Dr. Mohammad Ali Moni.

Tools and Machine-learning based projects

Tool name: NSF Grant Analyzer and Visualization Platform [GitHub]

- A Python-based tool for collecting, processing, and analyzing historical NSF award data.
- Created interactive visualizations to examine funding distributions across U.S. regions, research directorates, institutions, and scientific disciplines.
- Can be utilized to identify funding patterns and highly funded research topics.

Project Name: Early-Stage Cervical Cancer Detection Using Machine Learning [GitHub]

- A Python-based machine learning tool for predicting early-stage cervical cancer using clinical and demographic risk factors.
- Can be utilized to compare classification models and identify the most influential risk factors associated with cervical cancer diagnosis.

Project Name: Automated Tomato Detection Using YOLOv5 [GitHub]

– A deep learning-based project for detecting tomatoes in images using customized YOLOv5 models.

– Can be utilized for automated crop monitoring and real-time tomato detection in agricultural environments.

Awards and Achievements

– Scholarly Publication Recognition Award, Daffodil International University, awarded in recognition of publications in reputable indexed journals, 2020.

– Scholarly Publication Recognition Award, Daffodil International University, awarded in recognition of publications in reputable indexed journals, 2021.

– Nominated for Full Membership in Sigma Xi, The Scientific Research Honor Society, in recognition of scholarly achievements and contributions to scientific research, 2024

– ACM Professional Membership in recognition of authorship and contribution to ACM-BCB conference proceeding, 2026.

Academic Services

Reviewer

- IEEE Reviews in Biomedical Engineering Journal: 2022-Present
- Biology Methods and Protocols Journal: 2022-Present
- Healthcare Analytics Journal: 2023-Present
- Knowledge-Based Systems Journal: 2024-Present
- Gene Journal: 2024-Present
- Scientific Reports Journal: 2025-Present
- Molecular and Cellular Biochemistry Journal: 2025-Present
- BMC Cancer Journal: 2026-Present
- BMC Pharmacology and Toxicology Journal: 2026-Present
- Discover Genetics and Evolution Journal: 2026-Present

Presentations

– "CLOVER: A Cross-Cancer Learning Model Using Somatic Variant Data for Biomarker Recognition" at ACM-BCB-2026, Rende (Cosenza), Italy.

– "Drug Compound Prediction-Based Analysis of Cigarette Smoking in Pancreatic Cancer Patients: A Bioinformatics Study" at IEEE WIECON-ECE-2020, Bhubaneswar, India.

Reference

Dr. Suzan Arslanturk, Associate Professor, Dept. of Computer Science, Wayne State University

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Dr. Sara Masoud, Associate Professor, Dept. of Industrial and Systems Engineering, Wayne State University
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Dr. Amiangshu Bosu, Associate Professor, Dept. of Computer Science, Wayne State University
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